

DTSC 5810

Capstone/Practicum

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	None
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (28.9 / 100)	23.9 / 100

24

CURRENT AVG SCORE
across 6 job roles

77

POTENTIAL SCORE
with recommended additions

+80

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Students conduct approved research, critical review, or project work, presenting to a capstone committee for evaluation.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	28.9	94.8	+65.9	HIGH
AI Engineer	26.3	81.1	+54.8	HIGH
Machine Learning Engineer	24.7	72.4	+47.7	HIGH
ML Engineer	23.3	54.0	+30.7	HIGH
MLOps Engineer	20.6	58.0	+37.4	HIGH
Data Scientist	19.8	100.0	+80.2	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

System Design 3 of 6 roles

systems

LLMs & Generative AI 1 of 6 roles

storage

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1 HIGH Add System Design coverage** — required vocabulary in 3 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2 LOW Optional: LLMs & Generative AI** — would broaden appeal to one additional role. Candidate terms: storage.
- 3 LOW Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Applied Scientist

28.9 → 94.8 (+65.9 pts)

reasoning storage literature interdisciplinary technical data learning systems

AI Engineer

26.3 → 81.1 (+54.8 pts)

perception following data robotics learning build software technical

Machine Learning Engineer

24.7 → 72.4 (+47.7 pts)

data sets learning analytics tools exploratory business design

ML Engineer

23.3 → 54.0 (+30.7 pts)

data learning machine systems model world build product

MLOps Engineer

20.6 → 58.0 (+37.4 pts)

data model systems learning machine tools development

Data Scientist

19.8 → 100.0 (+80.2 pts)

data sets product analytics presentation build

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.

